SAMPLE

Earth's Sky: Aurora Australis

The Southern Lights

Aira Astralis

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Introduction

A quarter of the world's sky lies south of the tropics, yet the most famous light show on Earth is remembered almost entirely from its northern half. Ask most people to picture an aurora and they will describe green curtains over Norway or Alaska. The same phenomenon burns over the Southern Ocean every night of the year, wheeling above the ice of Antarctica, the storm-lashed islands of the subantarctic, and, on restless nights, the hills of Tasmania and the pastures of New Zealand's South Island. It is called the aurora australis, the southern lights, and its story is one of the great unfenced chapters of science.

The name is Latin for southern dawn, and it was given by analogy. The northern lights had been named first, when European natural philosophers of the seventeenth century reached for a word that could hold the pale, flickering glow they saw on winter horizons. When sailors finally confirmed that the far southern sky held a matching radiance, they borrowed the older name and turned it upside down. One light, two hemispheres, and a single physical engine driving both.

That engine is the subject of this book, but so is the asymmetry. The southern aurora is harder to reach, harder to watch, and harder to study than its northern twin. The ring it draws around the magnetic pole sits mostly over water, ice, and a continent with no permanent towns. For most of recorded history, the people best placed to see it were sealers, expedition crews, and penguins. As a result, the southern lights were confirmed by science centuries after the northern ones were described in medieval chronicles, and there are still gaps in what we know about how the two ovals behave differently under the same solar wind.

What follows is a complete tour of the phenomenon from the southern side of the globe. It begins with the age of discovery, when the first navigators to pierce the roaring forties brought home reports of light in the sky that many readers simply refused to believe. It explains the physics that turns charged particles from the Sun into green and red fire two hundred kilometers above the ice, and the peculiarities of Earth's southern magnetic field that give the southern oval a shape and temperament of its own. It follows the observers: the expedition crews who wintered in darkness by choice, the automated stations on the polar plateau that watch through blizzards and months of night, the satellite cameras that first showed humanity the full ring of light, and the photographers of Hobart and Invercargill who chase forecasts the way their northern cousins chase the lights over Tromso.

The book closes with the questions still open, and with the storms that occasionally remind the southern hemisphere that it lives under the same sky as everyone else. In May 2024, the strongest geomagnetic storm in two decades pushed the aurora over every Australian state and across New Zealand, and for a few nights the southern lights were seen by more people at once than in all the centuries before. If that event stirred your curiosity, this book is written for you. If you have already stood under a red glow in Tasmania and wondered what you were looking at, it is written for you too.

One light, two hemispheres. It is time the southern one had its story told properly.

Chapter 1: The Discovery of Aurora Australis

The northern lights were a documented part of European intellectual life for a thousand years before anyone in a position of scientific authority had seen their southern counterpart. Scandinavian chronicles described burning skies in the medieval period, and the term aurora borealis, northern dawn, is usually credited to Galileo in 1619. The southern lights enjoyed no such early patron. The great southern latitudes were simply unknown waters. The suspicion that a matching light might exist was logical, since the geometry of a magnetic planet suggested symmetry, but logic is not a sighting, and the far south kept its secrets behind the worst weather on Earth.

The breakthrough came from the most celebrated navigator of the age. On his second voyage of exploration, Captain James Cook took HMS Resolution and HMS Adventure south across the Antarctic Circle in the austral summer of 1772 and 1773, hunting for the terra australis incognita that geographers still drew on their maps. In February 1773, in the South Atlantic near the edge of the pack ice, officers on the Resolution recorded streams of pale light moving across the southern sky, and the expedition's accounts called it by the name it still carries: the aurora australis. The phrasing was cautious and the observations brief, but this is the entry that settles into the record as the first confirmed, credibly reported sighting of the southern lights by a scientific expedition. Cook's crew did not understand what they had seen. They understood, correctly, that it belonged to the same family as the lights their countrymen described from the north.

Skepticism did not die with Cook's report. For decades, the southern lights lived in an awkward category: witnessed by sealers, whalers, and naval men, but rarely by trained observers with instruments and the leisure to study what they saw. The whalemen of the southern ocean knew the glow well; a bright night of light in the sky was simply part of their working world, as unremarkable to them as a headland to a coastal pilot. The scientific community received such testimony politely and filed it under the plausible. What the subject needed was systematic observation, and the southern hemisphere was the hardest place on the planet to be systematic. It had no observatories, no winter stations, and no resident population at high latitude at all.

The nineteenth century filled the gap slowly. Ship after ship logged lights during southern voyages, and the scattered reports accumulated into a pattern no one could dismiss. Reports clustered around the same latitudes, appeared in the same months, and behaved in the same ways the northern lights behaved, flickering in curtains and arcs, often before or after disturbances in compass needles. The magneticians of the 1840s, who had begun measuring Earth's magnetic field with new precision, argued on theoretical grounds that any light tied to the magnetic poles must come in pairs. When the great magnetic observatories of the Antarctic were still decades in the future, the whalers' aurora was the only southern instrument anyone had, and it was already telling a consistent story.

Then came the era when science moved south to stay. The Belgian expedition aboard the Belgica was icebound in the Bellingshausen Sea through the winter of 1898, the first party ever to endure an Antarctic winter deliberately, and its crew watched the lights cycle overhead through the long night. The German expedition under Erich von Drygalski wintered beside the continent in 1902 and 1903, running a full geophysical program that included regular auroral watches. Robert Falcon Scott's Discovery expedition wintered at McMurdo Sound from 1902 to 1904, and its men recorded the lights through the polar night. One by one, the wintering stations of the heroic age turned the aurora australis from a rumor into a data set: scheduled watches, sketches, magnetic readings taken while the sky glowed, and the first disciplined attempts to measure the light rather than simply admire it.

The most romantic artifact of that era is a book. During the British Nimrod expedition of 1907 to 1909, Ernest Shackleton's party wintered at Cape Royds on Ross Island, and to occupy the dark months they set up a small printing press in a hut lit by seal oil and candle. The typesetting, printing, and binding were done by expedition members, with the covers made from the wooden packing cases that had carried their provisions. The volume they produced, titled Aurora Australis after the lights they watched outside the hut, was the first book ever made on the Antarctic continent. Perhaps a hundred copies were printed at the sign of the penguin, as the imprint page put it, and surviving copies are now treasured museum pieces. It is a fitting monument: the phenomenon that had been science's orphan for a century became, in that dim hut, the namesake of the first book of a continent.

The systematic era arrived with Douglas Mawson. The Australasian Antarctic Expedition of 1911 to 1914 established stations at Commonwealth Bay on the Antarctic coast and on subantarctic Macquarie Island, and made geophysics, including auroral observation, a central purpose of the enterprise. Mawson's men kept watch schedules, photographed the sky with all-sky cameras, and coordinated their records with magnetic instruments running in Australia, creating the first real ground network that bracketed the southern auroral zone. The Commonwealth Bay station sat at one of the windiest places on Earth, and the expedition's auroral logs are studded with entries written under conditions that would test any observer's commitment. Out of that program came the recognition that the southern lights were not a curiosity but a standing window onto the physics of the upper atmosphere, one that the southern hemisphere could claim as its own research inheritance.

The twentieth century added the eyes that no expedition could carry. Whalers gave way to research vessels, and eventually Australia named its flagship icebreaker for the phenomenon itself: the Aurora Australis, which served the national Antarctic program from 1990 to 2020, carrying generations of scientists south, before handing the route over to her successor. From deck level to satellite orbit, the discovery story arcs from a navigator's brief note in a logbook to a phenomenon now watched continuously by machines and by a global public. But the man who first put the name in writing stood on a wooden deck under the same light that glows tonight over the ice, and the distance between his few lines and our modern understanding is the story this book tells.